# Revolutionizing low orbital rockets: understanding if fiberglass is more efficient than carbon fiber for fins to yield higher acceleration

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## Introduction/Background

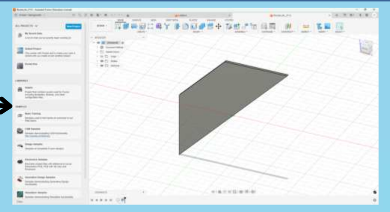
- Low orbital rockets are significant because they place satellites close to Earth for high-resolution imaging, real-time monitoring, and detailed studies in weather, environment, and mapping (Crisp, 2020)
- However, launching low-orbit rockets is no easy task, and one of the biggest challenges is making sure the rocket is stable during flight
- Stability is caused by fin flutter, which is when the fins on a rocket are bending during flight due to high velocity, material properties, and elasticity (Dinulovic, 2022)
- Instability could cause the rocket to change flight course and jeopardize the mission
- There is a correlation between stability and acceleration, where the acceleration will increase if the rocket is more stable
- Stability is caused by fin properties such as shape, material, and weight
- The fin material's weight, stiffness, and ability to withstand aerodynamic forces such as drag impact the rocket's stabilizing effect (Nakka, 2024)
- Acceleration is key because rockets need to achieve 17,500 mph to enter Earth's low orbital range
- Fiberglass and carbon fiber are materials that have polar opposite properties
- While carbon fiber is stronger, lighter, and very rigid, it is expensive, less impact resistant, and harder to fabricate, and vice versa for fiberglass

## Hypothesis & experimental factors

- **Research question:** Is fiberglass a more suitable material than carbon fiber for rocket fins to produce higher acceleration?
- **Hypothesis:** Fiberglass will be a more suitable material for rocket fins to yield higher acceleration than carbon fiber due to the greater elasticity and impact resistance fiberglass can withstand to combat fin flutter
- **Independent variable:** Material of rocket fins
- **Dependent variable:** Peak acceleration value in m/s^2
- **Constants:** Thicknesses being tested (1, 3, 5mm), model rocket kits, motors, altimeter/electronics
- **Control:** Peak acceleration values of carbon fiber since that is the industry standard

## Methodology

1. Trapezoidal-shaped fins designed on Fusion 360
2. The water jet application was used to cut out the fins from fiberglass and carbon fiber sheets of various thicknesses
3. Plastic cement was used to glue lower body tube parts together and glue that to upper body tube
4. Three fins were glued equidistant from each other to the lower body tube
5. 6 pieces of wadding tissue paper were placed at the body of the body tube to protect the altimeter and parachute from the heat of the ejection charge
6. The parachute was folded and tied to the nose cone; additionally, the shock cord was sent through the altimeter, and another was tied to the nose cone
7. A C6-5 motor was placed at the bottom of the rocket and fastened with a motor mount
8. Setting up the launch pad consisted of inserting each leg into the base of the pad, then connecting the three pieces of the launch rail, and then placing the rail onto the launch pad, and ensuring the rail is completely straight
9. Setting up the controller consisted of placing 4 triple-A batteries into the controller and testing to see if a red light would appear when launched
10. To launch, a starter and plug were inserted into the motor, and the rocket was placed on the launch rail
11. Two alligator clips were attached to the motor, and when the button was pressed on the controller, an electrical charge would heat the starter, igniting the motor



Water jet

Model rocket

Motors

Altimeter

Launch pad + Controller

## Data/Results

| Material | Thickness (mm | Peak acceleration (m/s^2) |
|---|---|---|
| Carbon fiber | 1 | 82.4 |
| Carbon fiber | 3 | 75.8 |
| Carbon fiber | 5 | 73.5 |
| fiberglass | 1 | 88.3 |
| fiberglass | 1 | 97.1 |
| fiberglass | 1 | 83.4 |
| fiberglass | 3 | 81.4 |
| fiberglass | 5 | 76.4 |

Figure 1 shows a data table of material & thickness (mm) vs. peak acceleration (m/s^2)

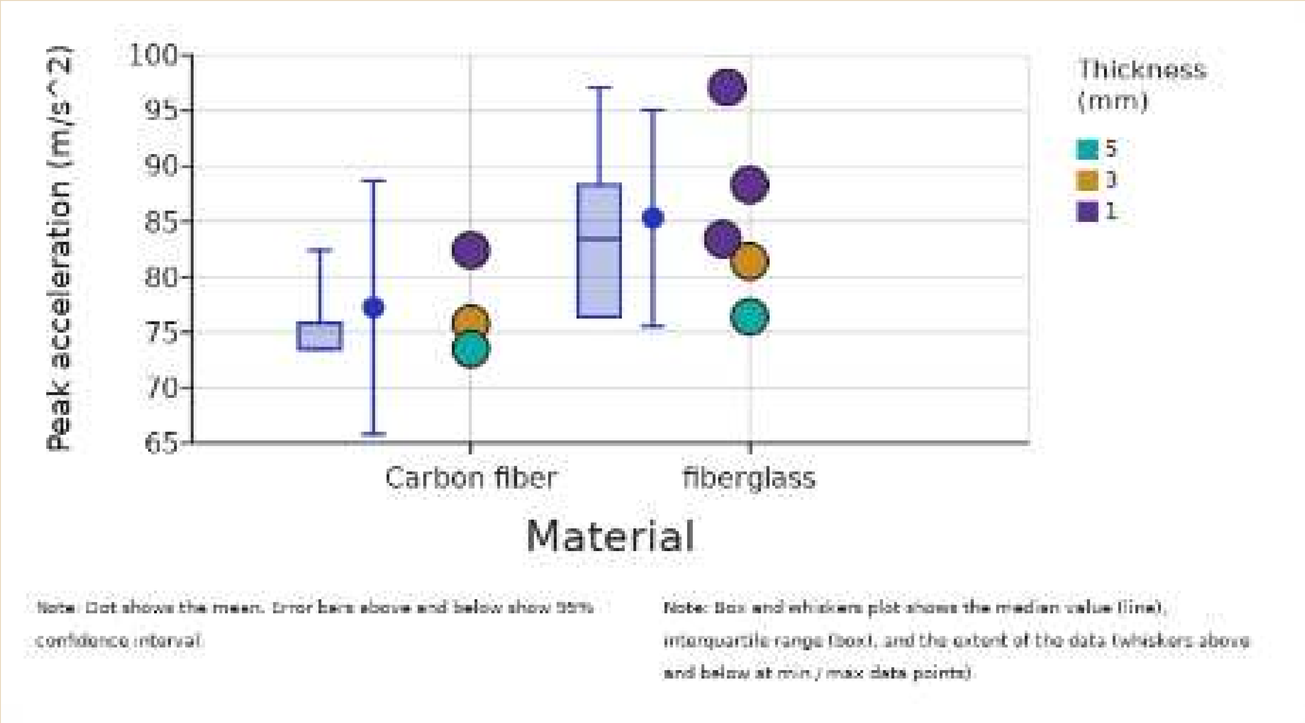



Figure 2 shows a graph of material & thickness (mm) vs. peak acceleration (m/s$^2$) with medians (box-and-whisker), and means with 95% confidence intervals

| Effect | Degrees of Freedom (df) | F-statistic (MS / MS residual) | P-value |
|---|---|---|---|
| Material (X) | 1 | 0.96 | 0.43 |
| Thickness (mm) (Z) | 2 | 1.7 | 0.38 |

Figure 3 shows the results of a 2-way ANOVA statistical test with the major IV being material and the secondary IV being thickness (mm)

## Conclusions

- Overall, fiberglass fins produced higher mean peak acceleration values than carbon fiber fins across all tested thicknesses, with the highest acceleration observed for 1 mm fiberglass fins (97.1 m/s²)
- Carbon fiber fins showed a general decrease in peak acceleration as thickness increased, while fiberglass maintained relatively higher performance at lower thicknesses
- These trends are evident in the acceleration graph, where fiberglass displays a slightly higher central tendency despite overlapping confidence intervals.
- A two-way ANOVA revealed that neither fin material ($F = 0.96$, $p = 0.43$) nor fin thickness ($F = 1.7$, $p = 0.38$) had a statistically significant effect on peak acceleration at the 0.05 significance level
- This indicates that although fiberglass demonstrated higher average acceleration, the observed differences were not large enough to be considered statistically significant given the variability and limited sample size
- Despite the lack of statistical significance, the results suggest fiberglass may offer practical advantages due to its elasticity and impact resistance, which could reduce fin flutter and energy loss during ascent.
- Future research should include a larger number of trials, additional fin geometries, and direct measurements of fin deformation to better evaluate how material properties influence rocket acceleration and stability.

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